

VEDANT PATIL

+91-8421340256 | vedantpatil1450@gmail.com | www.linkedin.com/in/vedant-patil-vp10 | github.com/vedpatil10

EDUCATION

K.J.Somaiya Institute of Technology, Mumbai

B.Tech in Electronics and Telecommunication and Honours in Data Science

CGPA: 9.47/10

Nov 2022 – May 2025

Trimurti Polytechnic

Diploma in Computer Engineering

Percentage: 80%

Dec 2020 – May 2022

CAREER OBJECTIVE

- Passionate software developer with expertise in Python, SQL, AI/ML and Full-Stack development, eager to apply technical skills in Data Science and software engineering to create innovative solutions.
- Experienced in building scalable applications and passionate about applying machine learning to solve real-world problems in collaborative, innovative environments.

WORK EXPERIENCE

Machine Learning Intern

Kids Galaxy, Mumbai

Nov 2023 – Jan 2024

- Developed an AI based chatbot to assist and guide students and help them to solve queries using chat services.

Data Science Intern

Spinnaker Analytics, Mumbai

Nov 2024 – Feb 2025

- Collaborated with the data team to clean, preprocess, and analyze large datasets, building ETL pipelines and creating visualizations in Matplotlib that improved business reporting efficiency by 25%. Assisted in developing and validating predictive models with scikit-learn, including feature engineering and tuning.

Full Stack Developer

IoT Cell, KJSIT, Mumbai

Jun 2024 – Sep 2024

- Built an event management platform with online registration & Razorpay payment integration, digitizing 100% of fest entries and reducing manual work.

Central Railway Intern

Mumbai

May 2023 – Jun 2023

- Monitored railway signal systems, documented findings, and collaborated across departments, gaining practical skills in data management, problem-solving, and large-scale workflow operations.

TECHNICAL SKILLS

Languages: Python, SQL, C++

Analytical: Machine Learning, Deep Learning (Neural Networks, CNN, LSTM), NLP, Computer Vision, Statistical Modelling

Frameworks & Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, OpenCV, Transformers (HuggingFace), Django, Flask

Web Development & Deployment: HTML5, CSS3, JavaScript (ES6+), ReactJS, Node.js, Express.js, PHP, WordPress, Streamlit

Databases: MongoDB, MySQL

Tools: Git, Linux, Jupyter Notebook, Google Colab

PROJECTS

Early Breast Cancer Detection using Machine Learning and Deep Learning (Python)

- Built an interactive system to classify mammogram images (*dataset: 9,686 images*) as *Benign* or *Malignant* using 8 ML/DL models (SVM, Random Forest, XGBoost, CNN, etc.).
- Achieved 90%+ testing accuracy across models, with XGBoost (93.4%), Random Forest (93.1%) and CNN (91.9%) performing best.
- Applied SMOTE method to handle class imbalance along image preprocessing and feature extraction to improve prediction quality.

Predictive Maintenance in Railways (Python, LSTM, TensorFlow, ReactJS)

- Built LSTM-based predictive maintenance system on 8,000+ multi-sensor time-series data, achieving 87.3% accuracy and 89.4% ROC AUC (10.5% better than traditional ML).
- Developed a ReactJS real-time dashboard with dynamic visualizations and failure probability alerts for live monitoring.

Auto Reply AI Chatbot (Python, Machine Learning)

- Developed an intelligent auto-reply chatbot that automates responses using machine learning models (BERT, Sentence Transformers) for sentiment analysis and intent recognition.
- The system processes incoming messages, analyses context and user intent, then automatically generates contextually appropriate responses with 85% accuracy in real-time.

AI Code Forge (ReactJS, Node.js, Express, OpenAI API)

- Engineered a production-style full-stack AI code generator supporting 12+ programming languages with real-time code generation, syntax highlighting, and an offline demo mode for testing.
- Optimized backend and UX with client-side generation history (20+ persistent sessions), secure CORS & Helmet integration, and a 1–2 s response time, boosting developer productivity.

CERTIFICATES

Data Base Management System: NPTEL (2023)

Linux OS: Spoken Tutorial (2024)

Python: Spoken Tutorial (2023)

Data Analytics and Visualization: Accenture (Forage-2024)

CO-CURRICULAR ACTIVITIES

- **CEO (IoT Cell):** Organized IoT events, workshops, and hackathons while managing team coordination and event execution.